

Week 9: Linear Transformation

INF1004 Linear Algebra

1 Introduction to Matrix Applications

Matrix algebra is a fundamental tool for applying mathematical concepts to engineering solutions. Key areas of application include:

- **Heat Transfer:** Modeling multi-layer heat distribution.
- **Kinematics & Robotics:** Analyzing movement and mechanics.
- **Image Processing:** Transformation and manipulation of digital images.
- **Machine Learning:** Neural networks and data science applications.
- **Electric Circuits:** Circuit analysis.

2 Matrix Notation of Linear Systems

A system of linear equations can be concisely written using matrices. For a system of 2 linear equations:

$$\begin{aligned} 2x + 3y &= 1 \\ x - 7y &= -14 \end{aligned} \tag{1}$$

This can be expressed in the form $A\mathbf{v} = \mathbf{b}$:

$$\underbrace{\begin{bmatrix} 2 & 3 \\ 1 & -7 \end{bmatrix}}_A \underbrace{\begin{bmatrix} x \\ y \end{bmatrix}}_{\mathbf{v}} = \underbrace{\begin{bmatrix} 1 \\ -14 \end{bmatrix}}_{\mathbf{b}} \tag{2}$$

where A is the matrix of coefficients, $\mathbf{v}$ is the vector of variables, and $\mathbf{b}$ is the vector of constants. In this context, matrix A 'transforms' vector $\mathbf{v}$ into vector $\mathbf{b}$.

3 Linear Transformations

3.1 Formal Definition

A linear transformation, $T : V \mapsto U$, takes an input vector space V and produces an output vector space U . It must satisfy the following properties for any vectors $\mathbf{v}_1, \mathbf{v}_2 \in V$ and any scalar $k \in \mathbb{R}$:

1. **Additivity:** $T(\mathbf{v}_1 + \mathbf{v}_2) = T(\mathbf{v}_1) + T(\mathbf{v}_2)$
2. **Homogeneity:** $T(k\mathbf{v}) = kT(\mathbf{v})$

A linear transformation is also known as a linear mapping or linear function. If we define $T(\mathbf{v}) = A\mathbf{v}$, it is a linear transformation because $A\mathbf{v}$ represents a linear combination of the columns of A .

3.2 Geometric Application: Image Transformation

Geometric shapes can be represented by matrices where coordinates are columns. Consider a triangle represented by matrix A :

$$A = \begin{bmatrix} 0 & 3 & 4 \\ 0 & 5 & 2 \end{bmatrix} \begin{matrix} \leftarrow x\text{-coordinates} \\ \leftarrow y\text{-coordinates} \end{matrix} \quad (3)$$

3.2.1 Translation

To move the triangle 3 units left and 1 unit down, we subtract values from the respective rows:

$$\begin{bmatrix} 0 & 3 & 4 \\ 0 & 5 & 2 \end{bmatrix} + \begin{bmatrix} -3 & -3 & -3 \\ -1 & -1 & -1 \end{bmatrix} = \begin{bmatrix} -3 & 0 & 1 \\ -1 & 4 & 1 \end{bmatrix} \quad (4)$$

3.2.2 Enlargement (Scaling)

To enlarge the triangle to twice its original perimeter, we multiply the matrix by the scalar 2:

$$2 \begin{bmatrix} 0 & 3 & 4 \\ 0 & 5 & 2 \end{bmatrix} = \begin{bmatrix} 0 & 6 & 8 \\ 0 & 10 & 4 \end{bmatrix} \quad (5)$$

3.2.3 Reflection

To reflect the triangle over the x-axis, we multiply by a transformation matrix B :

$$B = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \quad (6)$$

Calculating BA :

$$BA = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 0 & 3 & 4 \\ 0 & 5 & 2 \end{bmatrix} = \begin{bmatrix} 0 & 3 & 4 \\ 0 & -5 & -2 \end{bmatrix} \quad (7)$$

4 Inverse of a Matrix

A square $n \times n$ matrix A is **invertible** if there exists a square $n \times n$ matrix A^{-1} such that:

$$AA^{-1} = A^{-1}A = I \quad (8)$$

where I is the identity matrix. This is analogous to a reciprocal for numbers. The inverse allows us to reverse a linear transformation: $T^{-1}(\mathbf{u}) = A^{-1}\mathbf{u}$.

4.1 Calculating the Inverse

The general formula for an inverse relies on the determinant and the adjugate matrix (A_{adj}):

$$A^{-1} = \frac{1}{\det(A)} A_{adj} \quad (9)$$

where A_{adj} is the transpose of the cofactor matrix. If $\det(A) = 0$, the matrix is **singular** and not invertible.

4.1.1 2x2 Matrix Inverse

For $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, the inverse is given by:

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} \quad (10)$$

4.1.2 3x3 Matrix Inverse

For 3x3 matrices, the process is more tedious involving the full cofactor matrix. Example: For

$A = \begin{bmatrix} 1 & 0 & 1 \\ 2 & -2 & -1 \\ 3 & 0 & 0 \end{bmatrix}$, the determinant is 6, and the inverse is:

$$A^{-1} = \begin{bmatrix} 0 & 0 & 1/3 \\ -1/2 & -1/2 & 1/2 \\ 1 & 0 & -1/3 \end{bmatrix} \quad (11)$$

5 Solving Sets of Linear Equations

The practical importance of matrices lies in solving sets of linear equations $A\mathbf{x} = \mathbf{b}$.

5.1 Method 1: Inverse Matrix

If A^{-1} exists, the solution can be found by left-multiplying by the inverse:

$$\mathbf{x} = A^{-1}\mathbf{b} \quad (12)$$

In MATLAB, this is calculated as `x = inv(A)*b`.

5.2 Method 2: Cramer's Rule

For a system where $\det(A) \neq 0$, variables can be found using determinants. The determinant represents the area scale factor of the linear transformation in $\mathbb{R}^2$.

5.3 Method 3: Gaussian Elimination

This method avoids explicitly finding the inverse and uses row transformations on an **augmented matrix**. The MATLAB command `x = A\b` utilizes this method.

Procedure:

1. Write the equations as $Ax = b$.
2. Form an augmented matrix $[A|b]$.
3. Perform **Forward Elimination** (row transformations) to reach row echelon form. Allowed operations include:
 - Interchanging two rows.
 - Multiplying a row by a non-zero number.
 - Adding a multiple of one row to another.
4. Perform **Back Substitution** to solve for variables.

6 System Consistency and Solutions

When solving linear systems, three outcomes are possible:

Solution Type	Condition	Consistency
Unique Solution	All equations are linearly independent. Represents exactly one point of intersection.	Consistent
Infinite Solutions	Some equations are linearly dependent (e.g., a row of all zeros $0 = 0$). Represents infinite points of intersection.	Consistent
No Solution	Inconsistent system (e.g., a row yields $0 = k$ where $k \neq 0$). No point of intersection.	Inconsistent

Table 1: Summary of Solution Types